Project 4: Autostitching Photo Mosaics

A mosaic of the Berkeley campus!

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Part A.1: Shooting the Pictures

First, I captured a few sets of photographs of the same scene by rotating a camera while trying to keep its center of projection fixed. This results in photographs which can be transformed into one another via a projective transform.

Image 1 of the Grimes

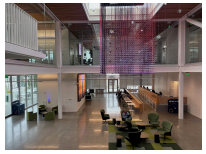

Image 2 of the Grimes

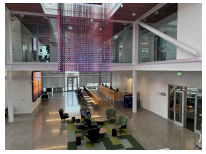

Image 3 of the Grimes

Engineering Center

Image 1 of the Berkeley campus

Engineering Center

Image 2 of the Berkeley campus

Engineering Center

Image 3 of the Berkeley campus

Engineering Center

Image 4 of the Berkeley campus

Image 5 of the Berkeley campus

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Part A.2: Recover Homographies

Next, I needed to find the transformation matrix H which could be applied to one image in order to warp it onto a shared projective plane as the second image. In this case, the transformation matrix is a homography matrix H with dimensions (3, 3) and 8 degrees of freedom (the lower right element is a scaling factor that can be set to 1).

A correspondence is a pair of pixel points (p', p) in two images that represent the same point in the 3D world.

Given a point p' in image 1 and a point p in image 2, left-multiplying the homogeneous coordinates of p by H maps it to p'. Dividing the resulting 3D vector by its last component to normalize it into a homogeneous coordinate with depth 1 gives:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (ax + by + c) \\ (dx + ey + f) \\ (gx + hy + 1) \end{bmatrix}$$

$$= \begin{bmatrix} \frac{(ax+by+c)}{(gx+hy+1)} \\ \\ \frac{(dx+ey+f)}{(gx+hy+1)} \\ \\ 1 \end{bmatrix}$$

Looking at just the non-homogeneous coordinates:

$$x' \cdot (gx + hy + 1) = ax + by + c,$$
$$y' \cdot (gx + hy + 1) = dx + ey + f$$

Rewriting the equations to isolate x' and y' on the left-hand side:

$$x' = ax + by + c - x' \cdot gx - x' \cdot hy$$
$$y' = dx + ey + f - y' \cdot gx - y' \cdot hy$$

Therefore, pair of points (p', p) gives us 2 equations. Since H has 8 degrees of freedom, we need at least 4 pairs of points to have a determined system of equations. Then, we will be able to set up a linear system of equations in the form of Ah=b, where h is a 8D vector holding the 8 entries of H.

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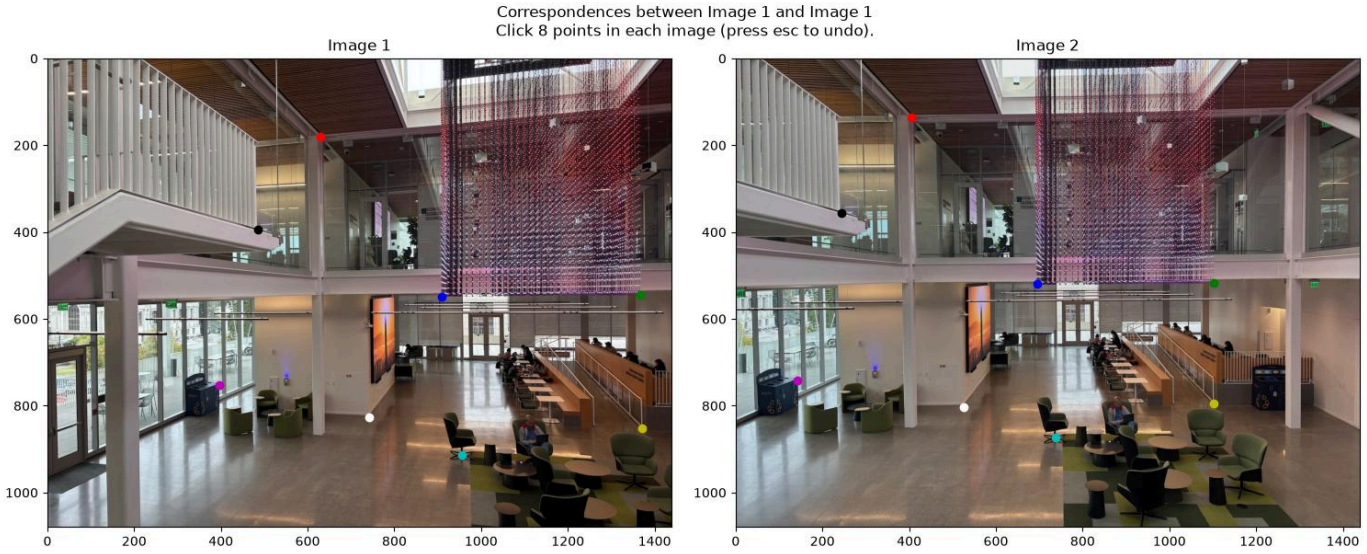
$$\begin{bmatrix} x_1' \\ y_1' \\ x_2' \\ y_2' \\ \vdots \\ x_n' \\ y_n' \end{bmatrix} = \begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x_1' \cdot x_1 & -x_1' \cdot y_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -y_1' \cdot x_1 & -y_1' \cdot y_1 \\ x_2 & y_2 & 1 & 0 & 0 & 0 & -x_2' \cdot x_2 & -x_2' \cdot y_2 \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -y_2' \cdot x_2 & -y_2' \cdot y_2 \\ & & & & \vdots & & & \\ x_n & y_n & 1 & 0 & 0 & 0 & -x_n' \cdot x_n & -x_n' \cdot y_n \\ 0 & 0 & 0 & x_n & y_n & 1 & -y_n' \cdot x_n & -y_n' \cdot y_n \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix}$$

The solution h will be unstable and prone to noise with only 4 points because an error of just a couple of pixels can result in very big changes in the recoevered homography. Thus & it's useful to instead use more than 4 point pairs, creating an over-determined system of equations, then find an h via least-squares.

In part A, I found correspondences between 2 images by implementing my own interactive widget that shows two images side-by-side and lets users select points in each image.

This is an example of how my correspondence-selection and homography algorithms work! This example used 8 correspondence point pairs, resulting in a system of 16 equations.



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Manually-selected correspondences between 2 images of the Grimes Engineering Center.

$$Ah = b$$

$$\begin{bmatrix} 9.09 \times 10^2 & 5.50 \times 10^2 & 1.00 & 0 & 0 & 0 & -6.32 \times 10^5 & -3.82 \times 10^5 \\ 0 & 0 & 0 & 9.09 \times 10^2 & 5.50 \times 10^2 & 1.00 & -4.72 \times 10^5 & -2.85 \times 10^5 \\ 1.37 \times 10^3 & 5.45 \times 10^2 & 1.00 & 0 & 0 & 0 & -1.51 \times 10^6 & -6.01 \times 10^5 \\ 0 & 0 & 0 & 1.37 \times 10^3 & 5.45 \times 10^2 & 1.00 & -7.08 \times 10^5 & -2.82 \times 10^5 \\ 6.30 \times 10^2 & 1.81 \times 10^2 & 1.00 & 0 & 0 & 0 & -2.55 \times 10^5 & -7.33 \times 10^4 \\ 0 & 0 & 0 & 6.30 \times 10^2 & 1.81 \times 10^2 & 1.00 & -8.59 \times 10^4 & -2.47 \times 10^4 \\ 9.56 \times 10^2 & 9.14 \times 10^2 & 1.00 & 0 & 0 & 0 & -7.05 \times 10^5 & -6.74 \times 10^5 \\ 0 & 0 & 0 & 9.56 \times 10^2 & 9.14 \times 10^2 & 1.00 & -8.35 \times 10^5 & -7.99 \times 10^5 \\ 3.96 \times 10^2 & 7.53 \times 10^2 & 1.00 & 0 & 0 & 0 & -5.64 \times 10^4 & -1.07 \times 10^5 \\ 0 & 0 & 0 & 3.96 \times 10^2 & 7.53 \times 10^2 & 1.00 & -2.94 \times 10^5 & -5.59 \times 10^5 \\ 1.37 \times 10^3 & 8.53 \times 10^2 & 1.00 & 0 & 0 & 0 & -1.51 \times 10^6 & -9.40 \times 10^5 \\ 0 & 0 & 0 & 1.37 \times 10^3 & 8.53 \times 10^2 & 1.00 & -1.09 \times 10^6 & -6.78 \times 10^5 \\ 4.84 \times 10^2 & 3.94 \times 10^2 & 1.00 & 0 & 0 & 0 & -1.18 \times 10^5 & -9.57 \times 10^4 \\ 0 & 0 & 0 & 4.84 \times 10^2 & 3.94 \times 10^2 & 1.00 & -1.72 \times 10^5 & -1.40 \times 10^5 \\ 7.42 \times 10^2 & 8.27 \times 10^2 & 1.00 & 0 & 0 & 0 & -3.89 \times 10^5 & -4.34 \times 10^5 \\ 0 & 0 & 0 & 7.42 \times 10^2 & 8.27 \times 10^2 & 1.00 & -5.96 \times 10^5 & -6.65 \times 10^5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix} =$$

Solving this system of equations via NumPy's linear algebra least-squares function gave this as the solution for h:

$$h = \begin{bmatrix} 1.35 \\ 9.35 \times 10^{-3} \\ -3.86 \times 10^2 \\ 1.24 \times 10^{-1} \\ 1.21 \\ -1.40 \times 10^2 \\ 2.37 \times 10^{-4} \\ 1.66 \times 10^{-5} \end{bmatrix}$$

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Finally, reshaping the vector h into a 3x3 matrix gives the homography matrix H:

$$H = \begin{bmatrix} 1.35 & 9.35 \times 10^{-3} & -3.86 \times 10^{2} \\ 1.24 \times 10^{-1} & 1.21 & -1.40 \times 10^{2} \\ 2.37 \times 10^{-4} & 1.66 \times 10^{-5} & 1.00 \end{bmatrix}$$

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Part A.3: Warp the Images

After recovering the homography, I warped one image onto another "reference" image's projective plane. I implemented this via inverse warping! First, I computed what the pixel positions of image 1's corners would be after warping. I used this as the bounding box to create an empty "canvas" of all zeros. Then, I iterated through all the pixel positions in the canvas. For each pixel position, I applied the inverse of the homography matrix to find which pixels in the source image it came from. Then, I "gathered" the values from the source pixels and set the output pixel to the gathered colors.

I implemented 2 interpolation methods from scratch:

Nearest neighbor interpolation: after inverting the output pixel coordinates, I rounded the resulting coordinates to find the nearest, valid source pixel position to gather a value from.

Bilinear interpolation: after inverting the output pixel coordinates, I computed the coordinates of the 4 neighboring pixels from the source image around the inverted coordinate, then used a weighted average of the neighbors' values in the output based on how close the inverted pixel coordinate is to each neighbor. To determine how much to weigh each neighbor's values, we can visualizethe neighbors as forming a 1 pixel by 1 pixel square. The inverted pixel coordinates fall somewhere inside this square (possibly on an edge of the square or on one of the neighbors themselves). The inverted pixel splits the 1x1 square into 4 sub-rectangles (or fewer than 4 sub-rectangles if the inverted pixel lies on an outer edge/corner). Then, each neighbor

should be assigned a weight proportional to the area of the sub-rectangle diagonal from it. For example, the neighbor above and to the left of the inverted pixel should be weighted by an amount proportional to the area of the bottom-right sub-rectangle.

For both interpolation methods, if the pixel in the warped output canvas was inverted to a coordinate that would be out-of-bounds in the source image, I treated that source pixel as having a value or weight of 0.

These two interpolation methods trade-off speed and quality. While nearest neighbors is algorithmically simpler and usually faster because it only requires "gathering" from at most one source pixel for every pixel in the output, bilinear interpolation can reduce image artifacts in the output since it more smoothly "blends"/combines colors from the source image in the output.

To test my homography and interpolation code, I applied them to pictures depicting known rectangular objects from non-front-on angles. This is known as "rectification": projecting images onto a common image plane (in this case, projecting images onto an plane such that the skewed rectangular objects look like rectangle again). Here are the results of rectifying a painting (*of a painting* in The National Gallery located in London!) and a fresco in Pompeii. I knew that in real-life, these paintings and frescoes were squares/rectangles with right angles and parallel edges. But because the images were taken from an angle, the paintings/frescoes look like skewed quadrilaterals. Rectifying these images involves pairing these images with a placeholder image of a normal rectangle to represent a projective plane that views the paintings/frescoes from front-on, computing a homography, and interpolating the colors from the original image to "unskew" the subject.

After rectification, the selected subject will now appear to have right angles and parallel lines!

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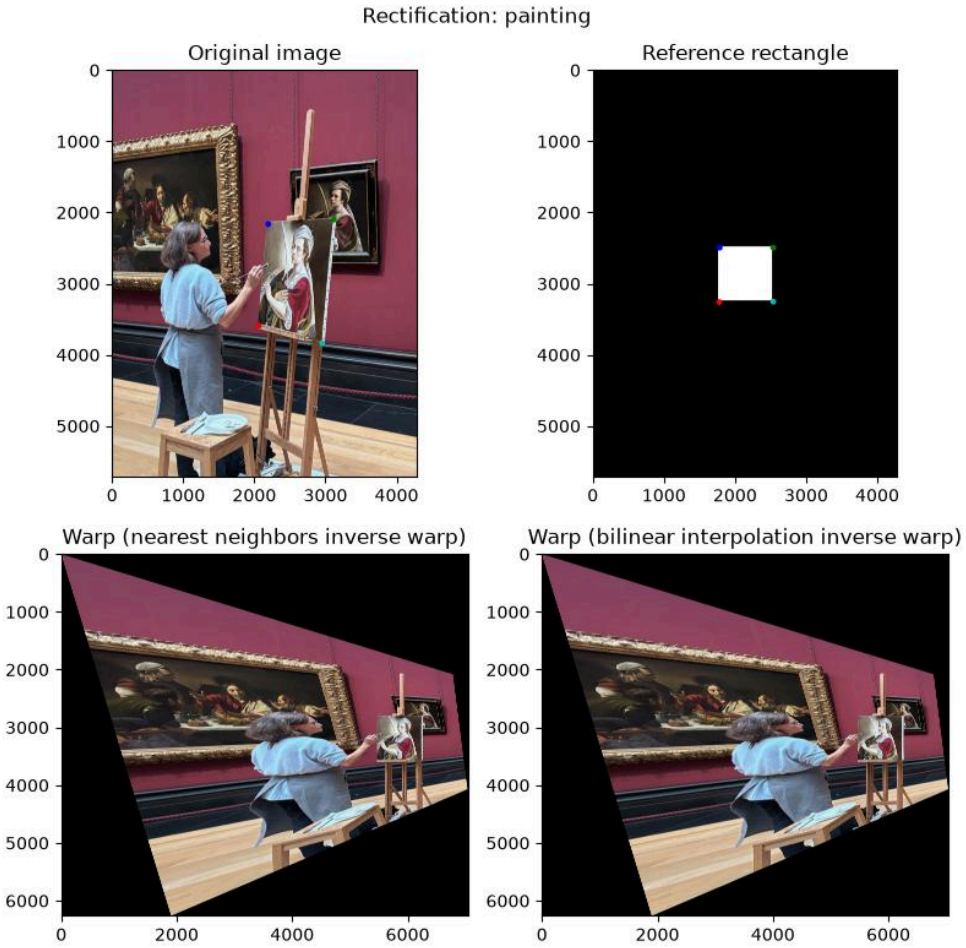

Rectifying a live painting of a painting. The top row shows the correspondences between corners of the painting and a reference square. The bottom row compares the results with both nearest neighbors interpolation and bilinear interpolation.

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painting.jpg's rectification using nearest neighbors interpolation.

painting.jpg's rectification using bilinear interpolation.

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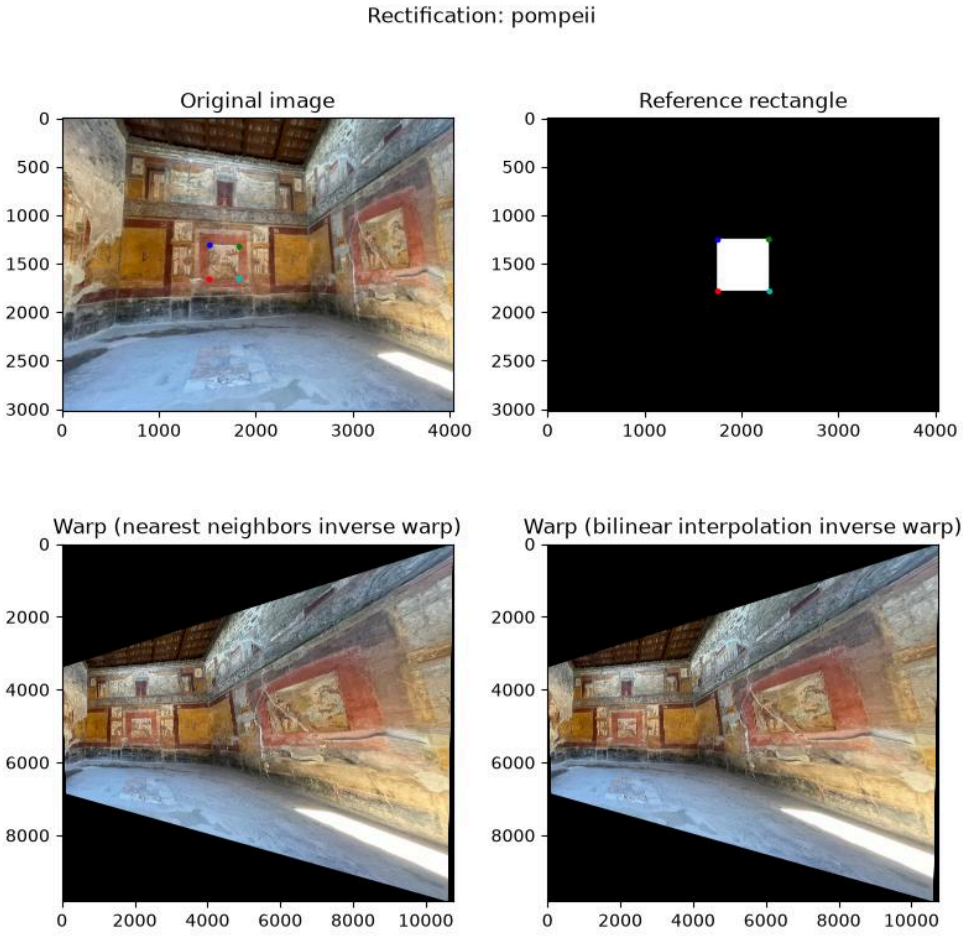

Rectifying a Pompeiian fresco.

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pompeii.jpg's rectification using nearest neighbors interpolation.

pompeii.jpg's rectification using bilinear interpolation.

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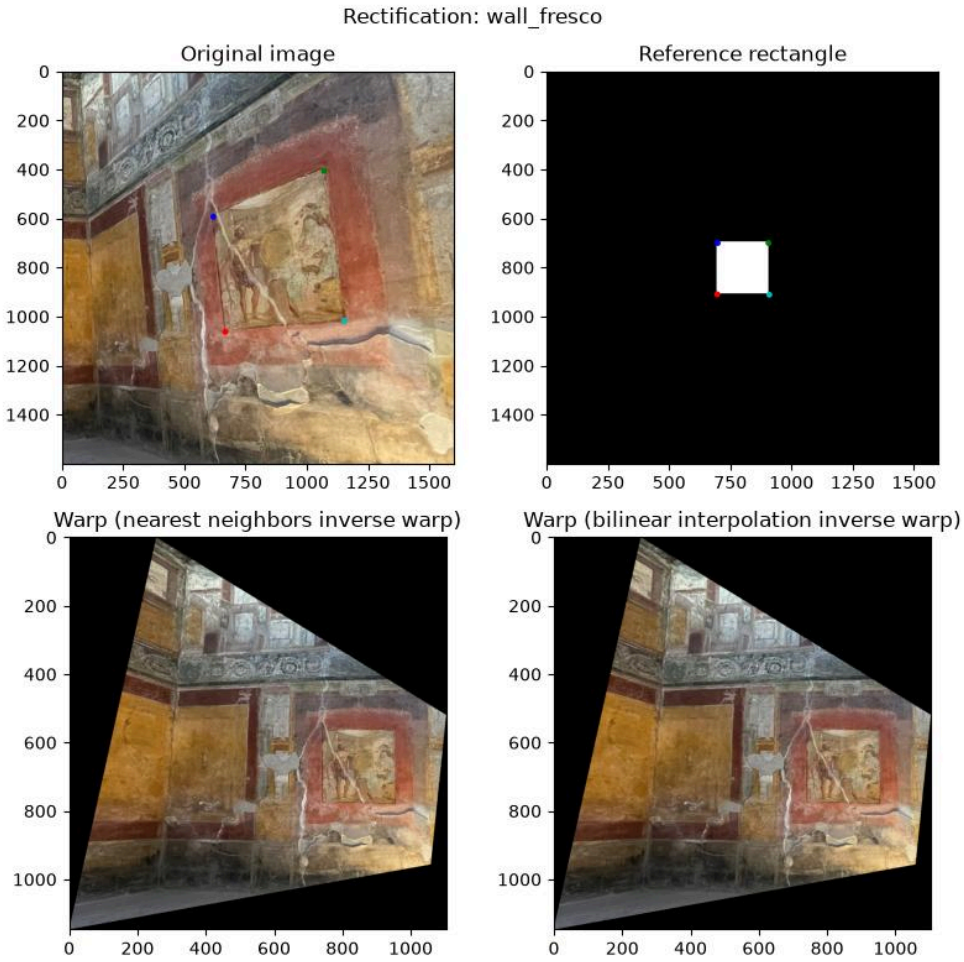

Rectifying the another Pompeiian fresco! This time, the fresco on the right wall in the picture.

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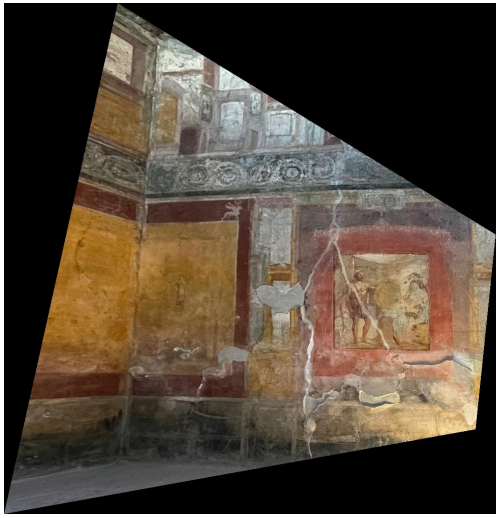
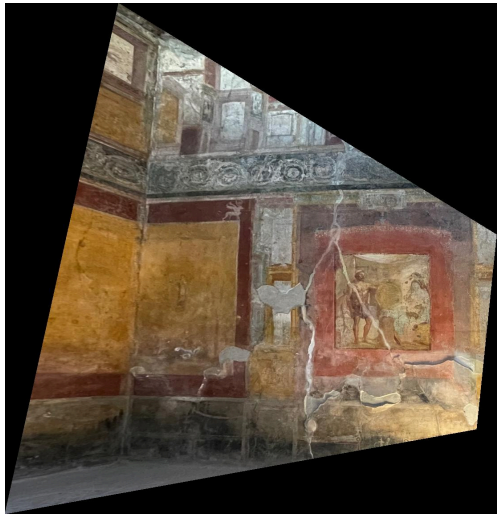


wall_fresco.jpg's rectification using nearest neighbors interpolation.



wall_fresco.jpg's rectification using bilinear interpolation.

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Part A.4: Blend the Images into a Mosaic

With all these puzzle pieces in place, we can now stitch multiple overlapping images into an image mosaic! In my implementation, I treated the middle image as the center of the mosaic, then transformed all the other images onto this center image's projective plane. To avoid edge effects, I used multi-resolution blending with Laplacian and Gaussian stacks.

1. Find correspondences between adjacent images and compute Homographies that map adjacent images to each other.

2. Compose the transforms together to find a homography for each image that maps it onto the shared, common projective plane (AKA the projective plane that the center image uses).

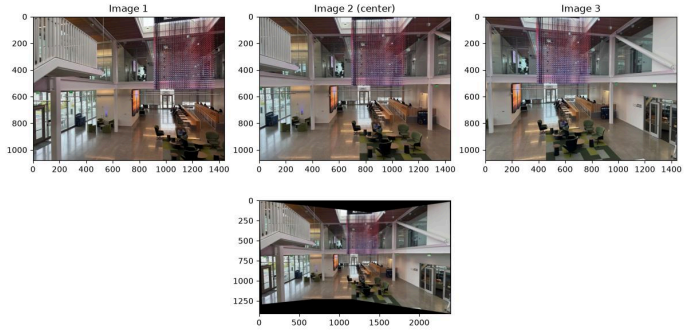
3. Compute the size that the shared canvas should have so that it fits all the images after they are tranformed. For each image, transform its 4 corners using the image-to-center homography. Then, find the minimum x, minimum y, maximum x, and maximum y

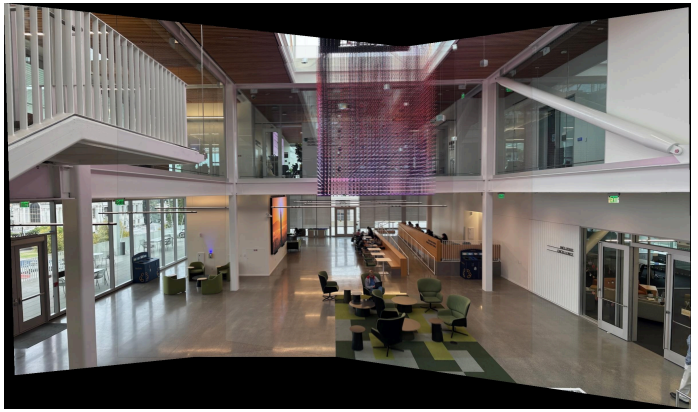
coordinates across all the warped corners. The shared canvas needs to be able to fit the full range of x, y coordinates; thus the dimensions of the final mosaic canvas should equal (maximum x - minimum x + 1, maximum y - minimum y + 1) pixels.

4. Warp each image onto its own copy of the canvas using the image-to-center homography and inverse interpolation. Using the same sized canvas helps simplify the multi-resolution blend step. Also, create a mask for each iamge. The mask should have the same shape as the canvas and have 1's where the warped image is located, and 0's elsewhere.

5. Blend all the image canvases together using multi-resolution blending! This involves creating Laplacian stacks of all the canvases containing a single warped image and Guassian stacks of all the masks. Then, merge each level by weighting each canvas's Laplacian at that level by its mask's Gaussian at that level. Finally, collapse (add) the blends from all the levels together.

Here are 3 photo mosaics I created from pictures I took on campus. There are still some blending artifacts like ghosting and blurriness. I attribute these to the fact that I took these photos by hand so the camera's center of projection may have slightly shifted between photos even though I tried my best to keep it fixed, and due to the fact that I manually selected corresponding keypoints between adjacent photos but the homography transform can be sensitive to even small pixel differences. Overall however, I think the resulting photo mosaics present a very *picturesque* look at everyday life at Cal :)





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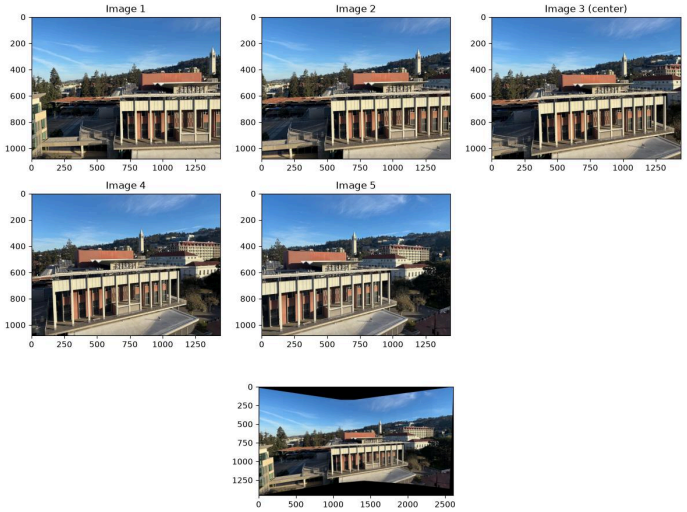
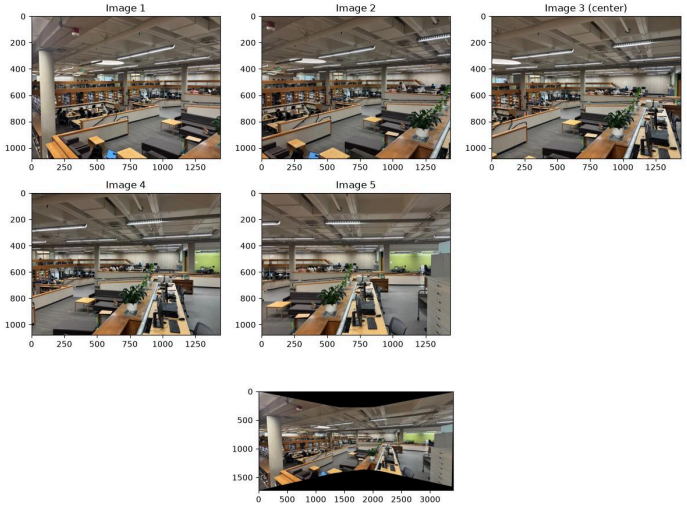
Images from inside the Grimes Engineering Center



Images of the Berkeley campus seen across from the Student Union.



Images from inside the Kresge Library.

Photo mosaic of the Grimes Engineering Center.

Photo mosaic of Berkeley campus.

Photo mosaic of Kresge Library.

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